

Signage

7.1 Overview

Existing signs at NAS Patuxent River vary significantly and use a wide range of materials, colors, and typefaces. An updated, unified signage system will instill a stronger sense of identity and provide more effective wayfinding throughout the entire base.

Webster OLF also exhibits a variety of signage styles and sizes. Most notably, street name signage is not present on any of the streets within the site. A special project to install street name identifiers throughout the fenceline is strongly recommended (See Chapter 9).

Signage at Webster OLF and NRC Solomons Island should conform to the requirements outlined

for NAS Patuxent River in order to present a consistent appearance throughout the complex. An updated signage system will benefit NAS Patuxent River in several tangible ways. First, a new signage system will be integral to both the experience and character of the base. It will contribute to the overall image of the complex, complementing the landscaping and site amenities. The signage system will have a significant presence on the site while providing organization and clarity through a unified set of signs.

Second, comprehensive signage will help bring order to the vehicular and pedestrian traffic by providing clear directions and information. Signage will help direct cars to desired routes and parking



Existing electronic signage at Main Base



Existing signage at Main Base



areas while also providing a measure of safety for pedestrians.

The general strategy for all signage in NDW shall be to adapt the signage currently in use at the Washington Navy Yard with site-specific modifications at each installation based on factors such as primary circulation patterns and building density. The signage at the Washington Navy Yard was designed prior to the publication of the IAG so additional modifications must be made to comply with recently established Navy signage standards. Some modifications will also be made to reduce cost and promote continuity throughout the region. In addition to conforming to Navy-wide standards, signage at NAS Patuxent River must also conform to the Manual on Uniform Traffic Control Devices (MUTCD) (2009) and the Maryland Manual on Uniform Traffic Devices (2006, rev. 2009).

7.2 Sign Components

In order to program, locate, and design the signage system, a hierarchy of elements must be developed as part of a sequence. This sequence begins when a visitor to NAS Patuxent River first encounters graphics at the adjacent street before entering the installation. From there, it progresses into the site, to drop-off and parking locations, and on to specific building, office, or tourist destinations. Upon exit, the sequence occurs in reverse. It is important to provide information during the entry and exit sequence in small amounts so that decisions can be made at appropriate intersections.

SIGN HIERARCHY

At NAS Patuxent River, a variety of sign types are required to provide a hierarchy. Signs which are intended to direct vehicles are different from those oriented to pedestrians. They should differ by type size and number of messages. However, these signs shall include common themes of color, type

styles, and nomenclature to create a cohesive system.

MESSAGE ORDER

Destinations on directional signage shall be listed in the order they will be encountered. The closest destination should be at the top of the list. As you approach a specific destination, the message hierarchy is modified.

TYPEFACE

The typeface for all signs on the Installation shall be Helvetica Medium, as dictated by the standards set in the Navy IAG. Helvetica Medium is one of the most widely used sans-serif font types in the world. It has an objective and functional style that allows for the clear presentation of information. Signs using this font will be easily read and understood. The simplicity of the font also allows for more letters per line, potentially reducing sign costs. Directional and identification signs shall use both upper and lowercase letters.

A B C D E F G H I J K L M N
O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n
o p q r s t u v w x y z

Helvetica Medium Typeface

NOMENCLATURE

A successful sign system relies on consistency and clarity of nomenclature.

- Destinations shall always be referred to with identical language.
- Destinations shall have short, functional names.
- Only the most universally understood acronyms shall be used. Where possible, spell out the names of buildings and destinations.
- Messages shall be short and provide as much information as possible.



MATERIALS

Messages shall be printed on panels composed of the same material as the sign. The panels shall be screwed to the body of the sign so they may be removed or changed if necessary.

All signs shall be composed of 0.090 in. thick, painted aluminum with screen-printed lettering. According to the Manual on Uniform Traffic Control Devices (MUTCD), 2009 Edition, all regulatory, warning, and guide signs, and object markers shall be retroreflective or illuminated to show the same colors by both day and night (Section 2A.07). The paint colors are as follows:

- Signage field: Blue, Pantone #19-4052 TPX
- Signage field for building-mounted identification signage: [Brown], Pantone #7532C (Brown is not a standard color name assigned to #7532C by Pantone, Inc.)
- Color band: Shall be determined according to Functional District (Figure 7.2 on the following page)

Sign posts can be square or round and composed of aluminum. Posts shall be painted [Brown], Pantone #7532C (Brown is not a standard color name assigned to #7532C by Pantone, Inc.), with the exception of street identification sign posts. According to the MUTCD, all post-mounted regulatory signs shall be mounted on breakaway or yielding posts if mounted within twelve feet of the roadway (Section 2A-19).

All signage paint shall be a Kynar 500© paint or like finish to ensure high durability and reduce the need for sign maintenance. Kynar 500© paints are resistant to color changes from UV light and a twenty year warranty protects against fading, chalking, and corrosion.

COLOR

In general, light text on a dark background is easier to read and allows for the use of reflective text on roadway signage. The background color

Signage field



Pantone 19-4052 TPX
Blue

Signage frame



Pantone 7532C
Brown

Functional District Color Bands



Administrative/Training
Pantone 18-6114 TPX
Myrtle



Airfield
Pantone 17-1019 TPX
Elmwood



Industrial
Pantone 17-4405 TPX
Monument



Personnel Support
Pantone 14-1031 TPX
Rattan

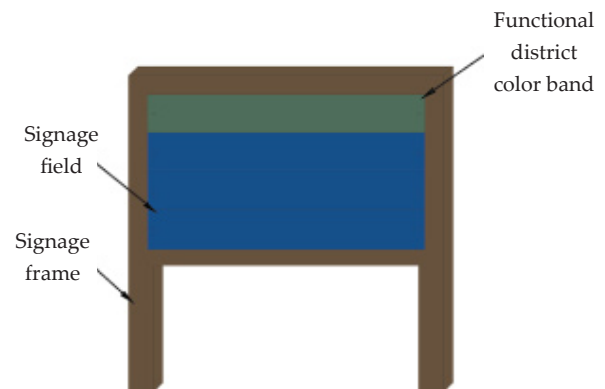


Figure 7.1 Illustration of IAG-Compliant Signage

for the signage field for all directional signs on the installation shall be blue, Pantone Color #19-4052 TPX, as dictated by the standards set in the IAG. Building-mounted identification signage will have a brown signage field. Street signage will continue to have a green field, as specified in the MUTCD. The letter color for all signage shall be white. Each sign shall also display a color band that corresponds to the Functional District in which the sign is located, with the exception of street signs. The colors are as follows:

- Administrative/Training: Myrtle, Pantone #18-6114 TPX



- Airfield: Elmwood, Pantone #17-1019 TPX
- Industrial: Monument, Pantone #17-4405 TPX
- Personnel Support: Rattan, Pantone #14-1031 TPX

The boundaries of the Functional Districts that will dictate the color used will be defined by an approximated future condition (approximately the year 2030) when the existing Functional Districts will be well-developed and occupy distinct areas of the Installation. These future Districts will be generalized to the most common activity type in that area. In the case that a building's function does not match the activity defined by its District,

the signage for that building will reflect the District, not the building's individual function.

Installation directories will be placed at entrances and along highly trafficked corridors and intersections that clarify the intent of the Functional District scheme for the Installation. The directories will display a map indicating the areas of the Installation in each Functional District and explain the coordination of the signage color band. The Functional District color band scheme is intended to aid a sailor on an unfamiliar installation in identifying his whereabouts so he may quickly and easily reach his destination.

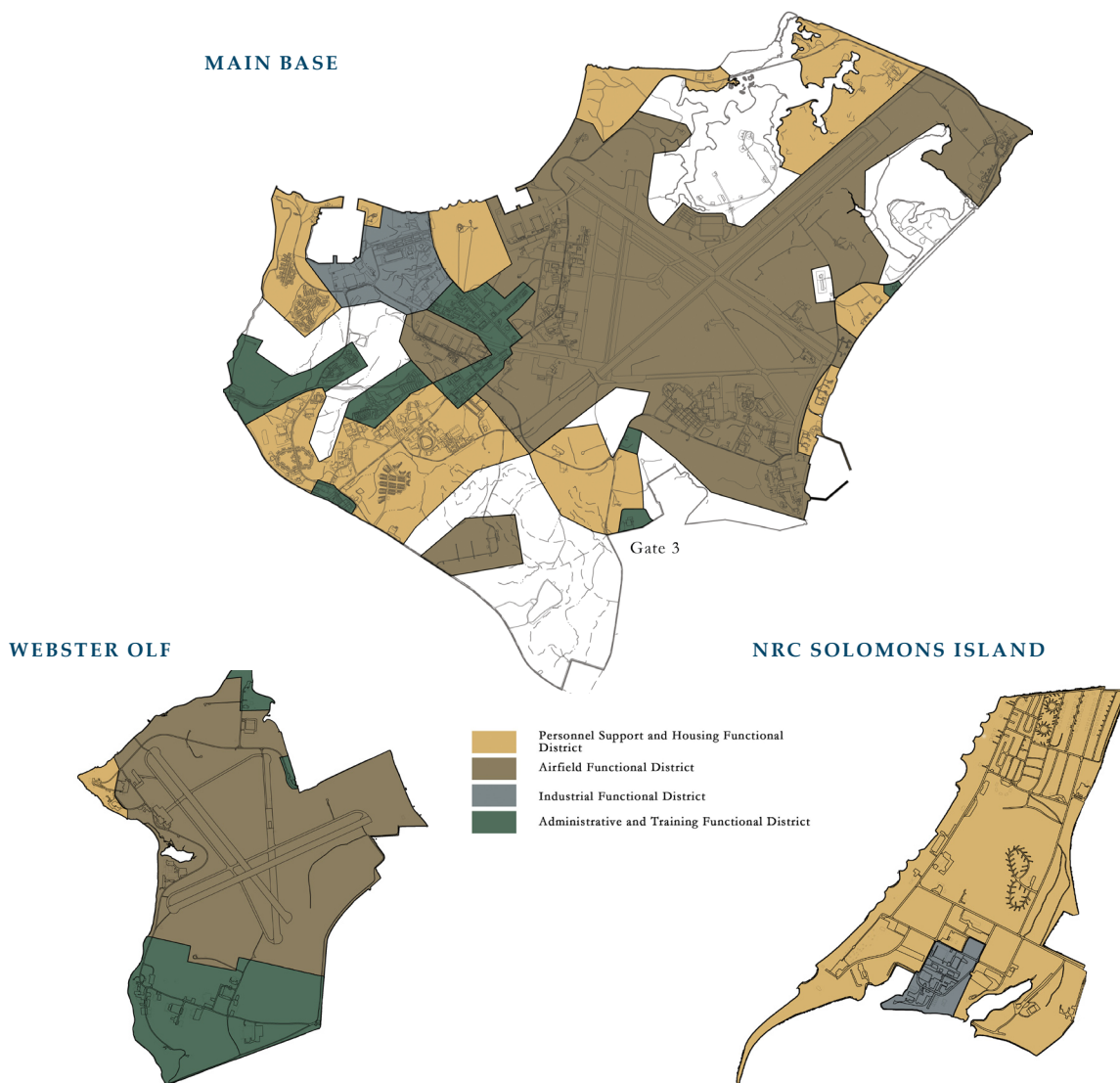


Figure 7.2 Functional Districts at NAS Patuxent River

7.3 Sign Types

Many types of signs are used to direct people around the installation. Below, eight types of signs are described.

SITE ENTRY IDENTIFIERS

The graphics at the project perimeter entrances are the starting point of the visitor experience and create a valuable first impression. They identify the installation from the surrounding roads and streets, and form an important symbolic role as the “gateway” to the installation. These signs should introduce the graphic theme of the site to the visitor, to provide them clues on what to look for as they progress.

The site entry identifiers shall be designed to respond to several key points:

- Locations should be chosen for maximum impact and visibility.
- The site entry identifiers shall introduce the Navy brand to the visitor, specifically by incorporating the Navy seal within each gate entry sign.
- The function of the site entry identifiers shall not be combined with directional messages; they should be kept solely as an entry marker.

- Signs should be illuminated in order to be visible at night and in low light.
- The design should be carefully coordinated with architectural and landscape features. The sign may be combined with these by mounting them on architectural or landscape walls, integrating with planting, and by using compatible materials.
- Primary and secondary entrances to the sites should have similar sign treatments at differing scales. The same basic design should be adapted at a smaller, appropriate size for secondary entrances.

STREET NAME IDENTIFICATION

All streets at the installation shall be marked by flag signs on individual poles located at intersections (see Figure 7.3 for example). Where applicable, street signs may be combined with existing light fixtures to reduce costs and provide ambient light for sign visibility. This is not permissible when the light fixtures are decorative. Street signs should follow these conventions:

- When used, sign poles shall be [Brown], Pantone #7532C, and constructed of the same aluminum used for other installation signage, and be of breakaway construction.



Example of site entry identifier at Gate 1 (Main Base)



- Signage field shall be green. This is a variation from general NDW standards which require a brown field, but necessary to ensure consistency with existing signage throughout the base, reduce driver confusion and comply with MUTCD standards.
- Lettering shall be white and retroreflective.
- Street name shall be displayed in 6 in. text for capital letters, 4.5 in. text for lower-case letters. Larger text is required due to the suburban nature of the installation and the 35 mph speed limit. For overhead street signage, a minimum of 12 in. text for capital letters and 9 in. text for lower-case letters shall be used, as recommended by the MUTCD.
- Overhead street name signage shall be used at major intersections within the installation where mast-arm traffic signals are present.

VEHICULAR DIRECTIONAL SIGNS

Vehicular directional signs provide information for those arriving and leaving the Installation by car (see Figure 7.4 for example). They are necessary to direct cars to the appropriate destinations.

- The viewer will require adequate time to read, understand, and react to the information provided. As the design speeds of roads increase, the driver's cone of vision decreases, so signs must be placed closer to the road, or the size of the lettering must be increased so it may be more easily read.
- Signs will have no more than 6 messages.
- Place signs a minimum of 6 ft. from the roadway and rotate the sign toward the road three degrees to reduce reflectivity.
- The bottom of the sign letters shall be a minimum of 30 in. above the ground for design

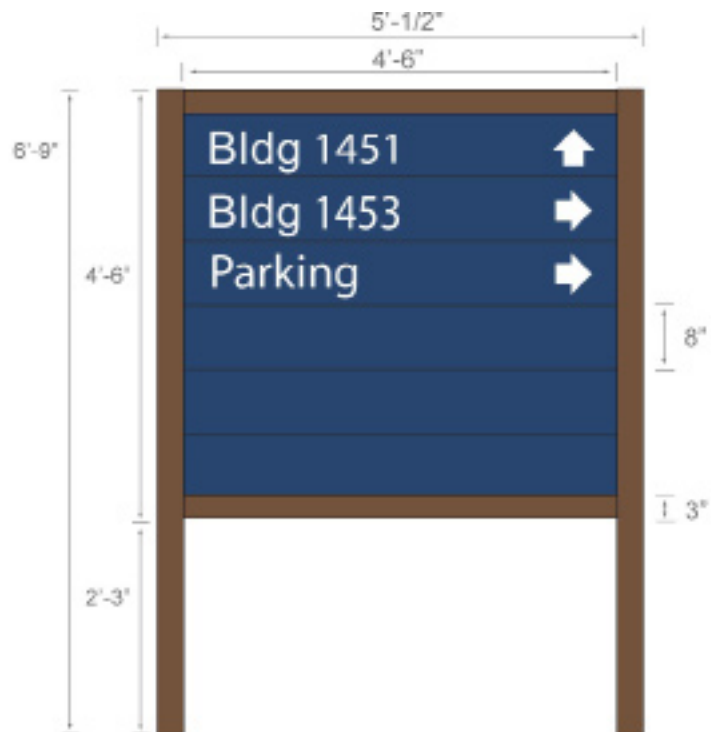
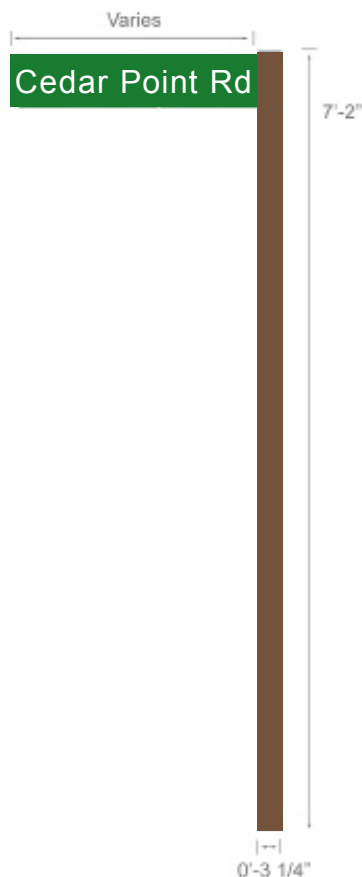


Figure 7.4 Vehicular Directional Sign

speeds below thirty miles per hour, and 36 in. for design speeds above thirty miles per hour.

- The letter size shall be designed in accordance with the following design speeds for roads:
- Less than twenty-five miles per hour: 3 in. lowercase letter height, which can be read from a distance of 150 ft.
- Twenty-five to thirty-five miles per hour: 4 in. lowercase letter height, which can be read from a distance of 200 ft.
- Thirty-six to forty-five miles per hour: 6 in. lowercase letter height, which can be read from a distance of 300 ft.
- Arrows shown on signs shall conform to a square proportion of two parts tall by two parts wide with the arrow shaft being one part tall by one part wide and the triangle point being two parts tall by one part wide (horizontal arrows). If the arrow is vertical, the triangle points shall be one part tall and two parts wide.
- Major directional signs shall be illuminated by floodlights, if possible. Reflective vinyl lettering shall also be used to ensure MUTCD requirements for visibility are met.

PEDESTRIAN WAYFINDING

Pedestrian directional signs will provide wayfinding information to specific destinations and buildings throughout the site.

- Pedestrian directional signs may have more messages than vehicular signs.
- Text height shall be a minimum of $\frac{3}{4}$ in. for capital letters, although 1 in. is preferred.
- Messages should direct pedestrians to building entrances.
- Pedestrian signs are viewed from a close distance, so better detailing and vandal resistance are necessary. Screen-printed messages are more durable than vinyl messages, which may be easily removed.

DIRECTORIES

Directories showing an overall site map and a list of destinations shall be designed for pedestrians and placed in highly circulated areas where the information is most applicable, such as visitor entry gates. Directories shall have changeable map artwork to allow for Installation building changes and additions. See Figure 7.5 on the following page for an example.

BUILDING IDENTIFICATION SIGNS

As dictated by Navy standards, all buildings on the Installation shall receive a building-mounted identification sign displaying the building's assigned number. See Figure 7.6 on the following page for an example.

- Building-mounted identification signs shall display only the building number. Building names shall not be included.
- For high-use, major destination buildings, an extra sign displaying the postal address shall be mounted below the building number sign.
- Buildings not considered high-use, major installation destinations, such as sheds and warehouses, shall receive building-mounted signage without the address.
- Building number shall be displayed in 6 in. or 8 in. text.
- Building address shall be displayed in 2 in. text.
- Signage field shall be [Brown], Pantone #7532C. Lettering shall be white.
- Signs shall be placed where they are most visible for all methods of circulation, preferably near the entrance.
- Canopies and awnings shall not be printed with building identification signage. Where identification is necessary, a building-mounted or free-standing sign shall be used.



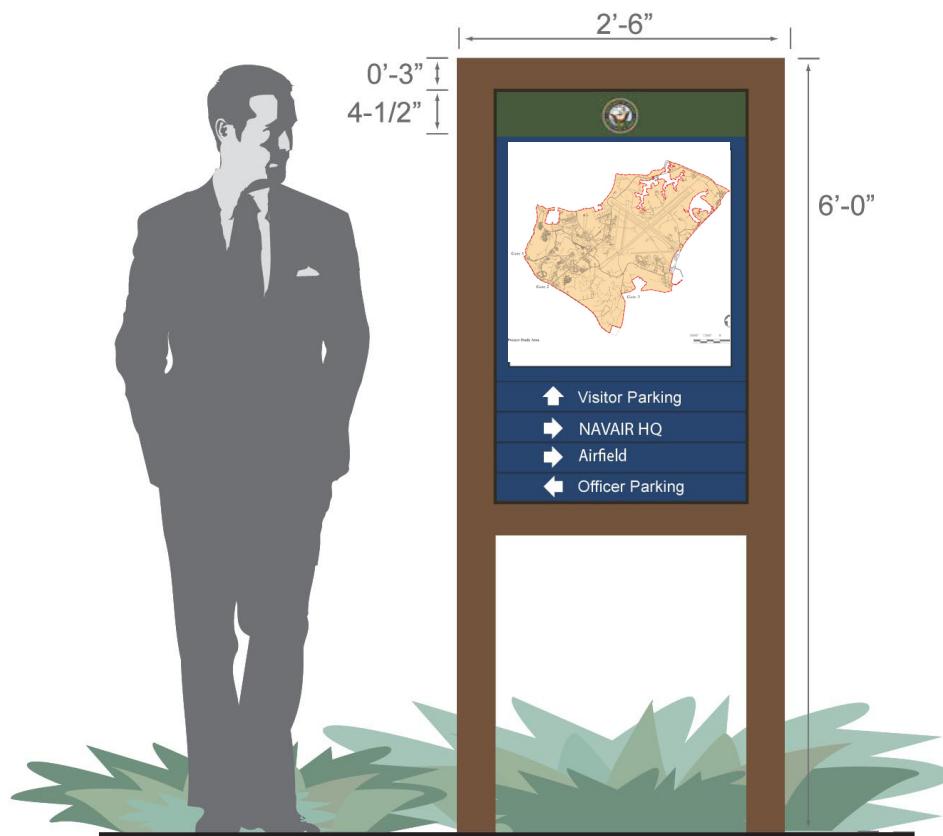


Figure 7.5 Pedestrian Directory (incorporating the functional district color band)

FREE-STANDING IDENTIFICATION SIGNS

In addition to a building-mounted identification sign, high-use buildings shall also receive a free-standing identification sign that displays the building's name, building number, and postal address. See Figure 7.7 for an example.

- Free-standing identification signs shall be consistent with all other signage on the Installation.
- Messages shall be at least 4 in. tall and 2 ft. above grade to allow for landscaping at the sign base.
- Signs shall be double-sided and placed perpendicular to the road for visibility from cars approaching from either direction.
- The free-standing identification signs shall include the building postal address. The postal address shall appear below the building name and number.
- Building address shall be displayed in 2 in. text.

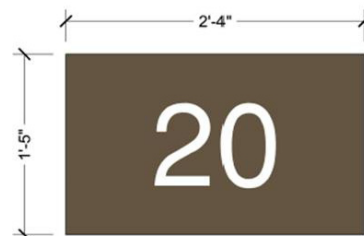


Figure 7.6 Building-Mounted Identification

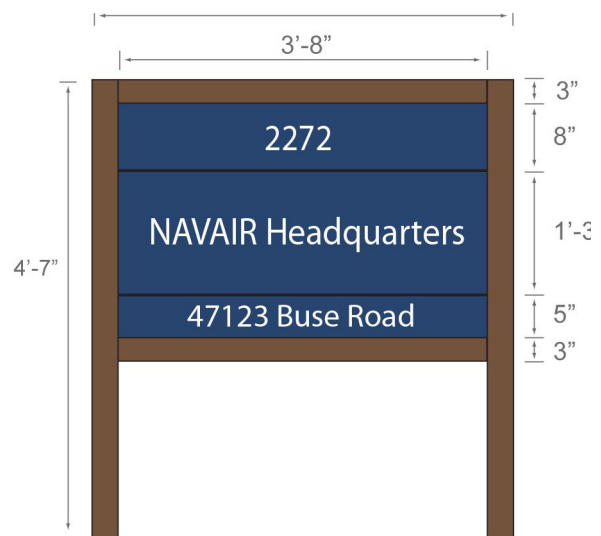


Figure 7.7 Free-Standing Building Identifier

REGULATORY SIGNS

Regulatory signs include standard traffic control signage, as well as messages specific to NAS Patuxent River, such as restricted areas, special warnings, and notices. Although they are primarily functional, the design of regulatory signs should relate to the rest of the system.

- For safety and legal reasons, critical traffic control signage (Stop, Yield) shall be placed by a qualified traffic consultant or civil engineer.
- Traffic signs shall use standard US Department of Transportation (DOT) sizes, shapes, and colors. For more information, reference the Manual on Uniform Traffic Control Devices (MUTCD).
- Wall-mounted signs may be used where appropriate. These may be used to provide visibility while maintaining sidewalk clearance and reducing sign damage.
- Pole-mounted regulatory signs shall use a pole to match that used for pole-mounted street identification signage. Poles shall be breakaway or yielding type poles if within the twelve-foot setback of the roadway.

COMMERCIAL SIGNAGE

Signs for commercial uses, such as retail stores, restaurants, and non-government functions, should complement the historic character of the installation. They should become a part of the overall composition, but not be allowed to dominate the non-commercial signage. Guidelines for commercial signs include:

- Messages on commercial signs shall be limited to the tenant's trade name or logo only.
- Appropriate sign types include panel signs with dimensional frames mounted flat to the wall surface, gold leaf on glass, or projecting blade signs.
- Use of nautical and historic-themed letters and graphics is encouraged. Natural materials such as painted or natural wood, natural metals,

brass or copper, wrought iron, tile, stained glass, and stone are encouraged for the signs and their details.

ELECTRONIC SIGNAGE

Electronic signage is a relatively new phenomenon at Navy installations. There are two large electronic signs currently located at the entrances to NAS Patuxent River Main Base and Webster OLF.

- Signage shall be rectangular, horizontal in orientation, with the display area generally comprising no more than 10 square feet. Pediments or other decorative elements that deviate from the rectangular form should be avoided unless they relate to the architecture of adjacent buildings.
- Signs shall rest on a vertical support that is 30 inches high at a minimum and at a maximum of 48 inches from the ground..
- The frame of the sign and the support shall be of brick construction.
- Use of electronic signs shall be limited to one per primary gate per installation, employing them only for the dissemination of key information that is subject to change.
- Consistent with the placement of vehicular directional signs, electronic signs shall be placed a minimum of six feet from the roadway.
- Light emitting diode (LED) signs should be employed rather than conventional electronic signs, as LED signs consume approximately one-tenth the amount of electricity required to power an incandescent sign. In addition, resolution is higher with LED technology.
- Luminance should be restricted at night to that which is necessary to maintain readability. Excessive luminance can contribute to light pollution at the installation.
- Electronic signage with streaming text or changing screens shall not be placed within the first 100 feet of entry gates, as they have the potential to increase driver distraction, thereby creating a safety hazard.



- Since electronic signage with streaming text or changing screens can increase driver distraction, such signs shall not be placed at complex or high volume intersections.
- Electronic signage should not be positioned where the information load is already high due to conventional signage.
- Electronic signage should not be located where drivers are frequently performing lane changing maneuvers.
- Double-sided signs should only be installed where the message can be clearly read from sides of the roadway without creating visual obstructions.
- All information, with the exception of the installation logo, shall be located within the electronic panel of the sign.

HISTORIC SIGNS

Interpretive signs can be used to provide information in historic areas. These lower, less intrusive signs shall be placed near significant historic features so as not to degrade the setting. See Figure 7.8 for an example.

- Sign height shall be 34 inches from the ground to the bottom of the front of the sign.

- Sign faces shall be 20 inches by 30 inches and made from fiberglass embedded graphic panels.
- Historic, interpretive graphics shall be specially designed and shall present an overall look consistent with existing signs.
- Posts and frames for the graphic panels shall be steel and painted Pantone 18-4022 TPX Bluesteel.
- Installation of these signs shall be performed by the sign supplier or to the manufacturer's specifications.
- Signs immediately adjacent to a roadway shall be upright panels. If a sign is not adjacent to a roadway, a tilted panel shall be used.



Example of upright historic panels at Main Base

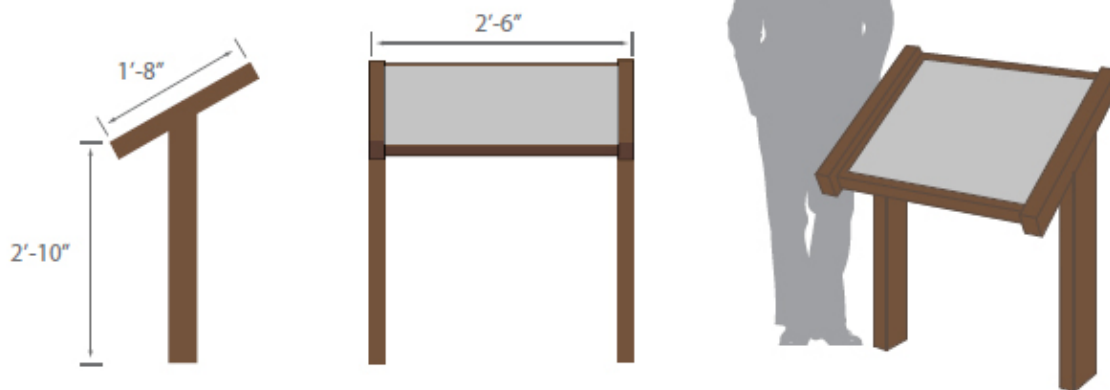


Figure 7.8 Interpretive Signage

ENVIRONMENTAL SIGNS

Environmental signs can be used to provide information about environmentally sensitive or unique areas. These lower, less intrusive signs shall be placed near significant environmental areas, habitats or zones so as not to degrade the setting.

- Sign height shall be 34 inches from the ground to the bottom of the front of the sign.
- Sign faces shall be 20 inches by 30 inches and made from fiberglass embedded graphic panels.
- Any graphics shall be specific and relevant.
- Posts and frames for the graphic panels shall be steel and painted Pantone 18-4022 TPX Bluesteel.
- Installation of these signs shall be performed by the sign manufacturer.
- Signs immediately adjacent to a roadway shall be upright panels. If a sign is not adjacent to a roadway, a tilted panel shall be used.

7.4 Procedures

The following guidelines are designed to assure that the necessary signage at NAS Patuxent River is coordinated, functionally effective, and visually attractive.

- All sign locations and appearance shall be approved by NAS Patuxent River Public Works Officer (PWO) prior to installation.
- Periodic inspection of signage throughout the base shall be performed to identify signs requiring maintenance.
- Deteriorated signs shall be repaired or replaced systematically based on above inspection report.
- All signs in a grouping or row shall be set an equal distance from the curb, sidewalk, or other related physical feature.
- Sign heights shall be appropriate to the intended viewer.
- In cases where signs will be mounted to walls or columns, signs shall be proportional to the architecture. Signs to be mounted to historic buildings or in historic districts require prior approval from the Cultural Resources Manager at the Public Works Department.
- No signs are to be mounted on decorative light standards.
- Consolidate multiple signs of the same type on a single post when possible.
- Posts should be installed plumb. Damaged, bent, and leaning posts shall be repaired systematically based on the scheduled inspection report.
- Mounting hardware shall be rust proof compatible metal when possible.

